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Employment

Tech Lead — **HubSpot** — **July 2024 – Present**

- Technical owner of the systems responsible for rendering all marketing e-mails, SMS, and WhatsApp messages sent by HubSpot, operating at company-critical scale and handling **over 10,000 requests per second** at peak;
- Designed and built a proof of concept for an AI-powered feature that personalises e-mails at the individual contact level. Defined the technical approach and core architecture; the solution evolved into a production-ready feature, was referenced in the keynote of HubSpot's Marketing Conference in 2025, and is **currently undergoing a patent application**;
- Led the design and implementation of several **AI-driven features** within the e-mail editor, including subject line and preview text generation using **LLMs** to **improve e-mail open rates** at scale;
- Provided technical direction across services, reviewed architectural changes, and collaborated closely with Product Management, UX, and Frontend teams to shape the roadmap and long-term system evolution, while supporting a distributed team of 5 engineers across the EU and UK;

Key Technologies: Java, SQL databases, Kafka-based systems, and internal platforms for LLMs, RAG, and LLM evaluation.

Senior Software Engineer — **Holistic AI** — **April 2023 – May 2024**

- Worked closely with Data Scientists to develop and deploy multiple **AI systems**, including setting up inference servers using Triton, deploying models, and building production-grade APIs;
- Designed and implemented a **RAG (Retrieval-Augmented Generation)** system using FAISS and LLMs to automatically extract information from client systems and generate AI-driven risk assessments;
- Reduced total execution time of a data extraction service from over **15 minutes** to under **2 minutes**, while lowering monthly operating costs from **thousands of pounds** to **a few hundred** by re-architecting the service and migrating to a Serverless architecture;

Key Technologies: Python, Java, TypeScript, React, with services deployed on GCP and Azure using Kubernetes, and AWS using a Serverless stack.

Software Engineer, Enterprise — **Meta** — **January 2022 – March 2023**

- Member of the team responsible for building internal supply chain software used across Meta;
- Led the implementation of Meta's internal approval workflow for requesting accessories and devices through the internal store. Delivered a configurable, UI-driven solution that significantly reduced engineering overhead and long-term maintenance effort, saving a projected **$1 million** in its first year;
- Onboarded and supported new engineers, acting as a key point of contact for system knowledge and team practices;

Key Technologies: Hack, Python, React, TypeScript, GraphQL.

Software Engineer — **NetEnt** — **January 2020 – December 2021**

- Built and scaled microservices for slot games, deployed across on-premise data centres and **GCP**, processing **hundreds of thousands** of game rounds **per minute**;
- Migrated a legacy monolithic system from an on-premise environment to **GCP**, followed by a transition to a microservices architecture;
- Addressed critical **latency issues** in time-sensitive reporting systems—key to maintaining the gambling licence—reducing response times from several seconds to a few **milliseconds**;

Key Technologies: Java, Scala, Kafka, PostgreSQL, and GCP services including GKE, Stackdriver, and Cloud SQL.

Software Development Engineer — **Amazon** — **January 2017 – December 2019**

- Developed high-scale backend services for **AFT (Amazon Fulfilment Technologies)** and customer-facing features for the Amazon Marketplace;
- Built **event-driven systems** using the AWS Serverless stack, including inventory and invoice matching algorithms, large-scale data ingestion pipelines, and data warehousing solutions;
- Delivered features used by **tens of millions of users** globally, meeting Amazon's strict **low-latency** requirements across search, checkout, payment instalments, and invoice generation workflows;
- Improved on-call operations by automating infrastructure provisioning and monitoring, using CloudFormation and Ruby tooling for documentation, alerts, and dashboards;
- Joined Amazon as an intern and transitioned to a full-time role in **May 2018**;

Key Technologies: Java, Python, JavaScript, and AWS services including Lambda, SQS, SNS, DynamoDB, and Redshift.

Education

B. Sc. Computer Science — **Universidade Mackenzie** — **July 2013 – July 2018**

- Conducted a research project that uncovered **one of the largest security breaches** in the Brazilian public transportation system, allowing free subway and bus access via smartphones. The findings were reported by several major newspapers; You can read the news coverage here;
- Final-year dissertation titled **Probability Models Applied to Football**, focused on implementing and evaluating statistical models to predict football match outcomes;

Exchange Programme — **The University of Sydney** — **July 2014 – July 2015**

- Awarded the **Science Without Borders scholarship**, a nationwide programme funding a one-year academic exchange and research placement;

Personal Projects

- **National Gallery Audioguide:** Built an AI-powered audioguide application capable of identifying paintings from photographs using **image similarity search** with a vector database, and generating narrated descriptions via multiple LLMs and TTS models. Live demo available here;
- **NFC Activity Monitor:** Developed an Android module that intercepts NFC calls at runtime to spoof NFC data. Repository;
- **Football Probability Models:** Implemented a collection of statistical models to estimate match outcomes and championship probabilities. Repository;